Fixing California's Existing Homes

Steve Mann is a HERS rater, Green Point rater, LEED AP, remodeler, and long-time software engineer.

In 1999, California adopted the first phase of its HERS program. Unlike HERS programs in the rest of the country, which generally focus on whole-house analysis, the California program focuses on field verification and testing of specific features identified in the state's Title 24 energy code. It is currently limited to new construction, building alterations and additions, and HVAC change-outs. All that is about to change with California HERS Phase II, which is currently undergoing public review by the California Energy Commission (CEC).

Old Homes, Bad Homes

The CEC is coming to grips with the fact that there are more than 12 million homes in California, over 70% of them built before 1978, the year Title 24 was enacted. The bulk of these older homes have serious deficiencies, things that home performance contractors have known about for a long time—low insulation levels, leaking ducts, single-pane windows, and leaky building envelopes, to name a few. The state is developing a methodology, called HERS Phase II, for evaluating these homes.

This methodology will be used to perform a whole-house analysis and assign a California HERS index to the house. This index can be used as the basis for financing an energy-efficient mortgage (EEM), comparing properties at time of sale, or giving homeowners a sense of how well their house performs. The index is very similar to the national HERS index. The major difference is that the California index uses time-dependent valuation to assign costs to electric and natural gas usage—energy used during peak demand periods, such as electricity for cooling in August, is more expensive than energy used during periods of low demand. A second difference is that the California index uses Title 24 to define its baseline building while the national HERS index uses the International Energy Conservation Code (IECC).

For a variety of reasons, including the need to respond to the CEC's efforts to define HERS Phase II, various organizations are developing programs, including software tools, for analyzing existing homes.

The first program is California Home Energy Efficiency Rating Services (CHEERS) for Existing Homes. CHEERS is one of three HERS providers in the state. Back in the mid-1990s, it had an existing-homes program, and a software application, for doing whole-house analysis. That program was suspended after a few years, because interest in EEMs dropped off. It has been redesigned and revived partly because there is a renewed interest in EEMs as a vehicle for financing upgrades to existing homes. The new program is aligning itself with the HERS Phase II regulations as they evolve. It uses two software applications—RateTool, an EnergyPro module, and the Registry Navigator, a Web-based program.

The second program is Build It Green's GreenPoint Rated Existing Homes (look for an article on this program in the Jan/Feb '09 issue). Although this program uses the California HERS index as its energy performance metric, it is an independent third-party rating system that provides a consumer label identifying the environmental performance of a home. It also helps homeowners to learn how to improve the energy efficiency and the indoor air quality (IAQ) of their homes, and how to save other valuable resources, such as water.

Both these programs use add-on modules for EnergyPro, one of the two state-certified Title 24 analysis software applications. Although EnergyPro is designed first and foremost for determining compliance with the California Energy Code, it has additional capabilities that make it useful for a variety of other needs. For instance, it has modules for calculating ASHRAE residential and commercial heating-and-cooling loads. It also has a module for calculating compliance for the residential energy efficiency federal tax credit.

CHEERS to the Rescue

When you first start EnergyPro with the RateTool module installed, you are presented with a blank screen. You can load an existing building, create a new building, or create a new building using a Wizard, which is the simplest way to get started. There is a Wizard specifically for the CHEERS Existing Homes program. Called the Home Rating Wizard, it steps you through a series of screens where you enter project information—location and orientation; default construction assemblies; floor, wall, and glazing areas; and specs for HVAC and water-heating equipment. This information defines your baseline house. You can also enter utility rate schedules so that the software can calculate dollar savings for each building improvement.

The last screen in the Wizard lets you enter a variety of modifications that you might want to make to the house (see Figure). There is a set of 12 predefined upgrades, including roof, wall, floor, and duct insulation; duct sealing; building sealing; HVAC, appliance, and window replacement; and more. You can enter the cost of, and the specifications for, each upgrade. Once you finish with the Wizard, you see the building tree.

The building tree is used to build a virtual building step by step. It is a rigidly defined hierarchy. From top to bottom, it consists of the following elements: building, plant (hot water source), HVAC system, heating and cooling zone, room, floor/wall/roof, and door/window. You can create very complex buildings using this hierarchy. You have the flexibility to define individual rooms, or to lump all similar rooms into one to make data entry simpler. You can also cut, copy, and paste sections of the tree, making it relatively easy to assemble a building with lots of parts. Each item at a lower level of the tree, such as walls and windows, has the default characteristics you specified when you used the Wizard. You can override those specifications by double-clicking on an item and entering other information.

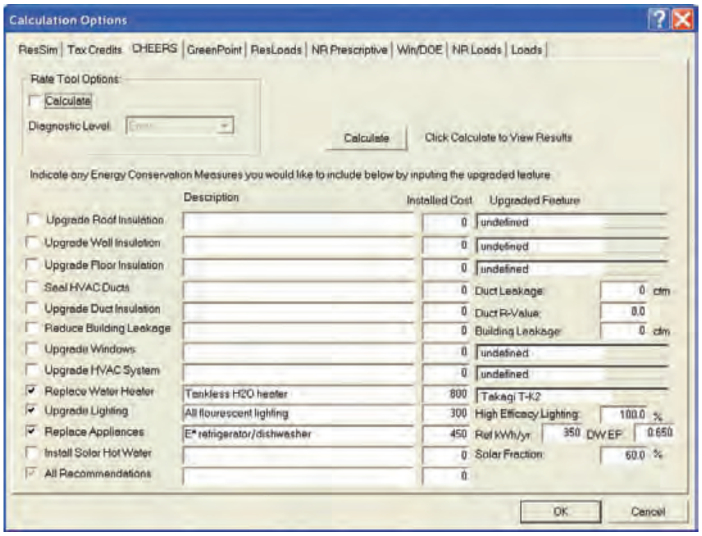

Figure. The Home Rating Wizard lets the user choose among 12 predefined building upgrades.

Once you have assembled the building, you select Calculate All from the CalcManager menu. The results that appear depend on the modules you are currently working with. For the CHEERS module, you get a window that lists the loads for heating, cooling, lighting, plugs, fans, pumps, and domestic hot water for the baseline reference house (HERS index of 100); for the building you are rating (without modifications); and for the building with each one of the modifications. It also shows changes to the HERS index for each upgrade. If you enter or select utility rate schedules, you also see annual dollar savings.

You can modify your building to your heart's content, trying different upgrade scenarios. Once you have a set of changes that you like and think your client will like, you can generate a CHEERS Rating Report.

CHEERS maintains a remote server, called the Registry, that contains all your company information and projects. In order to get an actual printed report from the CHEERS module, you have to download and install a Web-based application called the CHEERS Registry Navigator. For each project, you export the data from EnergyPro and import it into the Navigator. You can print draft reports and final reports. A final report gets stored in the CHEERS Registry and cannot be modified. There is a $25 charge for creating each final report, but you can create as many drafts as you like at no charge.

The first page of a draft report contains site information, recommended improvements, installation costs, annual savings, and the HERS scores for the existing and the improved versions of the home. The second page contains a more detailed breakdown of building characteristics, rates, and costs by categories. The report format is concise, informative, and easy to read.

Overall, the CHEERS EnergyPro module is straightforward and easy to use. Its accuracy and usefulness depend, of course, on the accuracy of the data you enter. Personally, I wish it would let you define custom modifications, but that feature is not currently available. CEC's Phase II regulations provide for what is called a custom approach to making recommendations. Hopefully, when that approach is fully scoped out, CHEERS will incorporate a bit more flexibility into its analyses.

There is a minor problem relating to utility rate schedules. The module includes a variety of schedules from all the state's major utilities. Unfortunately, it can be very tough to figure out which rate schedule to use. In theory, your utility bill should list the applicable rate schedule. If the CHEERS module doesn't have a rate schedule with an identical title, which happened to me for both gas and electric, you really have no easy way of identifying the appropriate schedule.

Fixing and Greening

Build It Green's GreenPoint Rated Existing Homes rating system evaluates a home in five environmental categories: community design, energy efficiency, IAQ, resource conservation, and water conservation. The Build It Green EnergyPro module uses the CEC's HERS Phase II draft guidelines for analyzing the energy performance of a building as part of this broader system.

You begin by entering all the data for the house—envelope, insulation, glazing, appliances, and so on. When all the data is entered, the program calculates the results and a window appears. It shows the loads for heating, cooling, lighting, hot water, and so on, as well as the California HERS index for the building. The application also calculates the energy performance points available to qualify for the GreenPoint Rated Existing Homes program.

The final step is printing a GPR-1R report. This resembles the standard Title 24 CF-1R report, with some Build It Green modifications. It includes a summary of the project, standard and proposed consumptions, and important GreenPoint Rated data: the California HERS index, the GreenPoint Rated energy performance points, and improvement percentages.

The Build It Green module calculates its energy performance score based on a target energy budget. The budget is based on parameters relative to the year the house was built. For newer homes, the parameters are based on California's Title 24 code. For older houses, the parameters are based on building practices at the time of construction. If the house uses less energy than its energy budget, it can qualify for the GreenPoint Rated Existing Homes program. This number is then factored into Build It Green's analysis of the home to determine its overall green rating.

software

Good to Go

Both the CHEERS and the Build It Green Existing Homes program were officially launched in spring 2008. By the time you read this, the programs should have the software and the administrative bugs worked out. They may also have incorporated some additional features into the modules.

There are different requirements for participating in each of these California-only programs. To enroll in the CHEERS program, you must first take the CHEERS three-day core HERS rater training class. CHEERS recommends, but doesn't require, that you take its HVAC and Building Envelope classes as well. Once you take the CHEERS class, and pass the exam, you are entitled to download and use the RateTool and the Registry Navigator software at no additional charge, except for the fee for each final report. There is also an annual fee of $125 to maintain your CHEERS certification. CHEERS membership includes telephone-based technical support and participation in its quality assurance programs.

The EnergyPro Build It Green module is available to the public from EnergySoft for $200. If you want to be an independent rater for the GreenPoint Rated program, there are some extra training requirements. You must first take and pass the GreenPoint Rater New Home training. You are then eligible to take the GreenPoint Rater Existing Home training. There are other prerequisites as well. Check Build It Green's Web site for a full list of requirements and the training schedule.

Although both these programs may seem, at first glance, to do much the same thing, they are actually quite complementary. The CHEERS program is designed for doing a whole-house analysis, making recommendations, and determining eligibility for an EEM. You can take the results of that program and fit them into the broader context of Build It Green's Whole House program to define a series of upgrades that can qualify the house for a GreenPoint Rated Existing Home label.

Using EnergyPro with both software modules makes it easy to integrate the two programs—you do the building takeoff once, enter the data once, and get reports for both. You could also use EnergyPro to do the Title 24 analysis, if required, and the federal tax credit analysis. Nothing beats one-stop shopping. He

For more information:

You can contact the author at steve@green-mann.com.

To find out more about EnergyPro software, including ordering and download information, go to www.energysoft.com.

For more on CHEERS for Existing Homes, go to www.cheers.org.

For more on Build It Green's GreenPoint Rated Existing Homes program, go to www.builditgreen.com.